

First name:

Last name:

Student ID:

Name and Student ID must be filled in for this page to be graded.

Problem 1. Use the exactness criteria to check if the following vector field is path independent in a simply connected domain:

$$\vec{F} = x^2y \, \hat{i} - 4xy^2 \, \hat{j} + 8z^2x \, \hat{k}.$$

[5 marks]

Solution.

$$\begin{aligned} \text{Check: } \vec{\nabla} \times \vec{F} &= \begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ \frac{\partial}{\partial x} & \frac{\partial}{\partial y} & \frac{\partial}{\partial z} \\ x^2y & -4xy^2 & 8z^2x \end{vmatrix} \\ &= \hat{i} \left(\frac{\partial}{\partial y} (8z^2x) - \frac{\partial}{\partial z} (-4xy^2) \right) \\ &\quad - \hat{j} \left(\frac{\partial}{\partial x} (8z^2x) - \frac{\partial}{\partial z} (x^2y) \right) \\ &\quad + \hat{k} \left(\frac{\partial}{\partial x} (-4xy^2) - \frac{\partial}{\partial y} (x^2y) \right) \\ &= \hat{i}(0) - \hat{j}(8z^2) + \hat{k}(-4y^2 - x^2) \neq \vec{0} \end{aligned}$$

What I am looking for:

- Exactness Criteria
as $\vec{\nabla} \times \vec{F} = \vec{0}$ or as
the system

$$\frac{\partial F_3}{\partial y} = \frac{\partial F_1}{\partial z}, \frac{\partial F_1}{\partial z} = \frac{\partial F_3}{\partial x},$$

$$\frac{\partial F_2}{\partial x} = \frac{\partial F_1}{\partial y}$$

- Correct calculations
- Not independent

As $\vec{\nabla} \times \vec{F} \neq \vec{0}$ in a simply connected domain D ,
thus $\vec{F}$ is not path independent.

□

First name:

Last name:

Student ID:

Name and Student ID must be filled in for this page to be graded.

Problem 2. Find $\oint_C \vec{F} \cdot d\vec{r}$ for the field $\vec{F} = 6y^2 \mathbf{i} + 2x - 2y^4 \mathbf{j}$, where C is the counter-clockwise boundary of the square region given by

$$R = \{(x, y) : -2 \leq x \leq 2, -2 \leq y \leq 2\}.$$

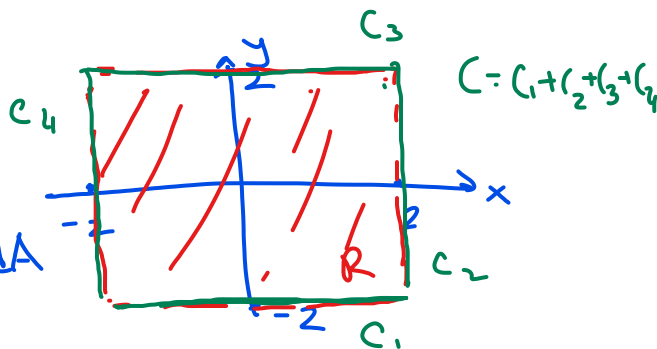
Hint: There is no restriction on how to compute the result, so any valid method will be accepted.

[5 marks]

Solution.

Using Green's Theorem:

$$\oint_C \vec{F} \cdot d\vec{r} = \iint_R \left(\frac{\partial}{\partial x}(2x - 2y^4) - \frac{\partial}{\partial y}(6y^2) \right) dA$$



What I'm looking for: $= \int_{-2}^2 \int_{-2}^2 (2 - 12y) dx dy$

- Either Green's Thm or using piecewise line integrals $= \int_{-2}^2 [2x - 12yx]_{-2}^2 dy = \int_{-2}^2 ((2(2) - 12y(2)) - (2(-2) - 12y(-2))) dy$

- Correct calculations $= \int_{-2}^2 (8 - 48y) dy = [8y - 24y^2]_{-2}^2$

- final answer is 32.

$$= (8(2) - 24(2)^2) - (8(-2) - 24(-2)^2) = (16 - 96) - (-16 - 96) = 16 - 96 + 16 + 96 = 32.$$

Alternate solution:

$$\oint_C \vec{F} \cdot d\vec{r} = \int_{C_1} \vec{F} \cdot d\vec{r} + \int_{C_2} \vec{F} \cdot d\vec{r} + \int_{C_3} \vec{F} \cdot d\vec{r} + \int_{C_4} \vec{F} \cdot d\vec{r}$$

$$= \int_{-2}^2 6t^2 dt + \int_{-2}^2 4 - 2t^4 dt + \int_{-2}^2 -6t^2 dt + \int_{-2}^2 4 + 2(-t)^4 dt = 2(4t) \Big|_{-2}^2 = 32$$

$$C_1: x(t) = t, y = -2, -2 \leq t \leq 2$$

$$C_2: x(t) = 2, y(t) = t, -2 \leq t \leq 2$$

$$C_3: x(t) = -t, y(t) = 2, -2 \leq t \leq 2$$

$$C_4: x(t) = -2, y(t) = -t, -2 \leq t \leq 2$$